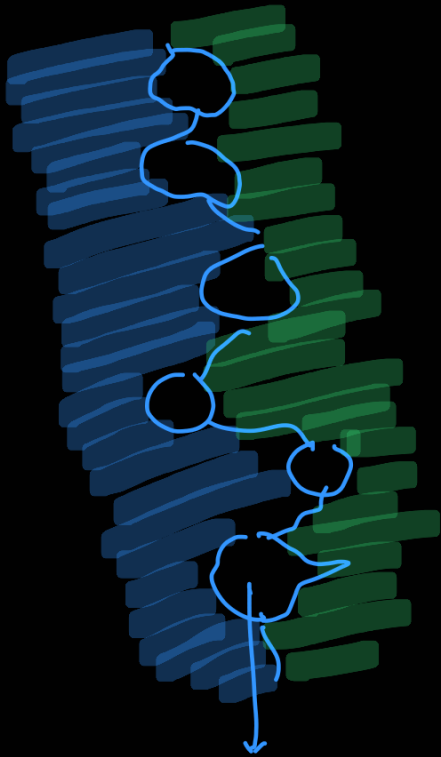


TYPES OF PDF

15/01/24

- Linear
- Non-Linear
- Quasi-linear
- Homogeneous
- Non-homogeneous.

BOUNDARY CONDITION AT INTERFACE



$$q'_s = -k_1 \frac{dT_1}{dx} = h_c (T_1 - T_2) = -k_2 \frac{dT_2}{dx}$$

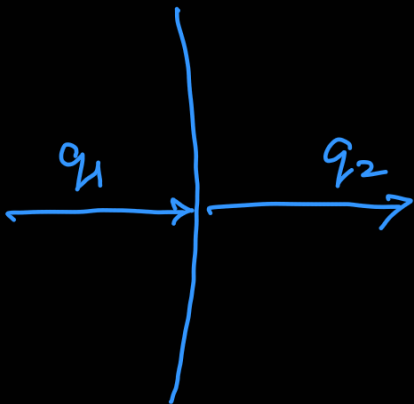
h_c : contact-conductance at interface

$$-k_1 \frac{dT_1}{dx} = -k_2 \frac{dT_2}{dx}$$

When there is no contact imperfections.

microscopic imperfections at interface

BC AT SOLID INTERFACE: RELATIVE MOTION



$$q_1 + \mu p V - q_2 = 0$$

↓
extra term due to friction.

NON-LINEARITY

→ Governing Eqn (Radiation, T dependent) parameters

→ Boundary conditions (free convection)

ORTHOTROPIC SOLIDS

Different heat conduction parameters in different directions.

Parallel to orthogonal directions.

$$\begin{bmatrix} k_{11} & k_{12} & k_{13} \\ k_{21} & k_{22} & k_{23} \\ k_{31} & k_{32} & k_{33} \end{bmatrix} \longrightarrow \begin{bmatrix} k_{11} & 0 & 0 \\ 0 & k_{22} & 0 \\ 0 & 0 & k_{33} \end{bmatrix}$$

$$k_{11} \frac{\partial^2 T}{\partial x^2} + k_{22} \frac{\partial^2 T}{\partial y^2} + k_{33} \frac{\partial^2 T}{\partial z^2} + \underset{\substack{|| \\ q'''}}{q(x, y, z, t)} = \rho c \frac{\partial T}{\partial t}$$

(volumetric heat gen)

Define stretched co-ordinates axes,

$$x_1 = x \left(\frac{k}{k_{11}} \right)^{1/2} \quad y_1 = y \left(\frac{k}{k_{22}} \right)^{1/2} \quad z_1 = z \left(\frac{k}{k_{33}} \right)^{1/2}$$

$$\frac{\partial^2 T}{\partial x^2} = \frac{\partial^2 T}{\partial x_1^2} \left(\frac{k}{k_{11}} \right) \quad \frac{\partial^2 T}{\partial y^2} = \frac{\partial^2 T}{\partial y_1^2} \left(\frac{k}{k_{22}} \right) \quad \dots$$

$$\boxed{k \left(\frac{\partial^2 T}{\partial x_1^2} + \frac{\partial^2 T}{\partial y_1^2} + \frac{\partial^2 T}{\partial z_1^2} \right) + q(x_1, y_1, z_1, t) = \rho c \frac{\partial T}{\partial t}}$$

[Similar to isotropic solid]

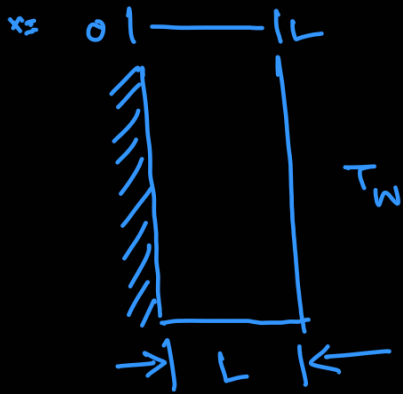
$$dx dy dz = \frac{(k_{11} k_{22} k_{33})^{1/2}}{k^{3/2}} dx_1 dy_1 dz_1$$

$$\text{If } dx dy dz = dx_1 dy_1 dz_1,$$

$$\boxed{k = (k_{11} k_{22} k_{33})^{1/3}}$$

$$K = (K_{11} \ K_{22} \ K_{33})$$

EXAMPLE 1:



Gov Eqn: $\frac{\partial}{\partial x} \left(K(T) \frac{\partial T}{\partial x} \right) + q''' = 0$

BCs: $\frac{\partial T}{\partial x} \Big|_{x=0} = 0$

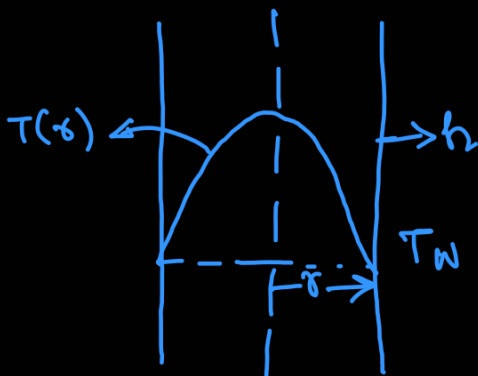
$$T \Big|_{x=L} = T_w$$

$$\frac{K_r}{K} \cdot \frac{\partial \theta}{\partial x} = 0$$

$$\theta(L) = \frac{1}{K_r} \cdot \int_{T_r}^{T_w} K(T) dT$$

PROBLEM 1:

Find q''' in a rod that will produce a centerline temp of 200°C for following conditions.



$r = 1\text{cm}$, $T_0 = 350^\circ\text{C}$, $K = \frac{3167}{T + 273}$

Calculate surface heat flux.

SOLN: Boundary conditions,

@ $r=R$, $T = T_w$ @ $r=0$, $\frac{\partial T}{\partial r} = 0$

Assume steady state. $\frac{\partial T}{\partial t} = 0$

$$\frac{1}{r} \frac{d}{dr} \left[r K(T) \frac{dT}{dr} \right] + q''' = 0$$

$$\theta(r) = \frac{1}{K_w} \int_{T_w}^{T(r)} K(T') dT'$$

$$K_w = K(T_w)$$

$$\frac{dT}{dr} = \frac{dT}{d\theta} \cdot \frac{d\theta}{dr} = \frac{K_w}{K(T)} \cdot \frac{d\theta}{dr}$$

$$\frac{1}{r} \frac{d}{dr} \left[r \frac{d\theta}{dr} \right] + \frac{q}{k_w} = 0$$

$$\left. \frac{d\theta}{dr} \right|_{r=0} = 0$$

$$\theta(r) = \frac{\dot{q} R^2}{4k_w} \left[1 - \left(\frac{r}{R} \right)^2 \right]$$

$$\theta(R) = 0$$

$$\int_{T_w}^{T(r)} k(T') dT' = \frac{\dot{q} R^2}{4} \left[1 - \left(\frac{r}{R} \right)^2 \right]$$

Placing $k(T') = \frac{3167}{T' + 273}$, $\int_{T_w}^{T(r)} \frac{3167}{T' + 273} dT' = \frac{\dot{q} R^2}{4} \left[1 - \frac{r^2}{R^2} \right]$

At $r=0$, $T = 2000^\circ\text{C}$,

$$\int_{T_w}^{T(0)} \frac{3167}{T' + 273} dT' = \frac{\dot{q} R^2}{4}$$

$$3167 \ln \left(\frac{2273}{623} \right) = \frac{(0.01)^2}{4} \times \dot{q}$$

$$\boxed{\dot{q} = 1.64 \times 10^8 \text{ W/m}^3}$$

Surface heat flux,

$$\boxed{q'' = \frac{\dot{q} \pi R^2 L}{2 \pi R L} = \frac{\dot{q} R}{2} = 8.2 \times 10^6 \text{ W/m}^2}$$

NON-FOURIER HEAT CONDUCTION

$$\rho c \frac{\partial T}{\partial t} + \nabla \cdot q'' - q''' = 0$$

$$q(r, t) = -k \nabla T(r, t)$$

$$\frac{1}{\alpha} \frac{\partial T}{\partial t} = \nabla^2 T + \frac{q'''}{k} \quad [\text{Parabolic PDE}]$$

Assumption: Heat propagation occurs at infinite

speed.

New relation: $q(r,t) + \tau_0 \frac{\partial q(r,t)}{\partial t} = -k \nabla T(r,t)$
 τ_0 : relaxation time, material property

Corresponding Heat conduction equation,

$$\left[\frac{1}{\alpha} \frac{\partial T}{\partial t} + \frac{\tau_0}{\alpha} \frac{\partial^2 T}{\partial t^2} = \nabla^2 T + \frac{1}{k} \left(q''' + \tau_0 \frac{\partial q'''}{\partial t} \right) \right]$$

Equation is of hyperbolic type. Combined diffusion and wave-like behavior, predicts finite speed for heat propagation.

Standard Wave : $\frac{\partial^2 q}{\partial t^2} = c^2 \left[\frac{\partial^2 q}{\partial x^2} + \frac{\partial^2 q}{\partial y^2} + \frac{\partial^2 q}{\partial z^2} \right]$
Eqn

$$\boxed{V_{cv} = \sqrt{\frac{k}{\rho c \tau_0}} = \sqrt{\frac{\alpha}{\tau_0}}} \quad (\text{Speed of heat})$$

SINGLE PHASE LAGGING MODEL

$$q(r,t) + \tau_0 \frac{\partial q(r,t)}{\partial t} = -k \nabla T(r,t)$$

It is a first order Taylor series expansion of a more general eqn.

$$\boxed{q(r,t + \tau_0) = -k \nabla T(r,t)}$$

[Single phase
Lagging model]

There is a finite built up time for temp gradient to heat flux.

For solids, $\tau_0 \in [10^{-14}, 10^{-10}]$

For gases, $\tau_0 \in [10^{-10}, 10^{-8}]$

For some biological systems, τ_0 can go up to 100 s.

DUAL PHASE LAGGING MODEL

$$q(r, t + \tau_0) = -k \nabla T(r, t + \tau_0)$$

Temp gradient at a point r of the material at time $t + \tau_0$.

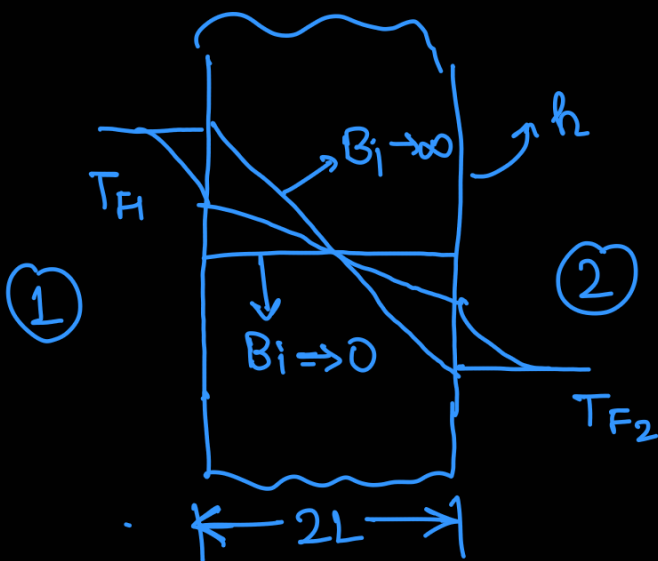
Performing Taylor series expansion,

$$q(r, t) + \tau_0 \frac{\partial q(r, t)}{\partial t} = -k \left\{ \nabla T(r, t) + \tau_0 \frac{\partial}{\partial t} (\nabla T(r, t)) \right\}$$

Substituting q in energy eqn,

$$\frac{1}{\alpha} \frac{\partial T}{\partial t} + \frac{\tau_0}{\alpha} \frac{\partial^2 T}{\partial t^2} = \nabla^2 T + \tau_0 \frac{\partial}{\partial t} (\nabla^2 T) + \frac{1}{k} \left(q''' + \tau_0 \frac{\partial q''}{\partial t} \right)$$

BIOT NUMBER: TWO LIMITING CASES



$$q = \frac{T_{F1} - T_{F2}}{\sum R_i}$$

$$\sum R_i = \frac{2}{hA} + \frac{2L}{kA}$$

$$Bi = \frac{\text{Internal res}}{\text{surface res}} = \frac{hL}{k}$$

(Biot No)

CASE 2: $Bi \rightarrow 0$

CASE 1: $h_1 \rightarrow \infty$

$$\sum R_1 \approx \frac{2L}{KA}$$

CASE 2: $h_2 \rightarrow \infty$

$$\sum R_1 \approx \frac{2}{hA}$$

ELIMINATE HEAT LOSS FROM ONE SIDE

